

Assignment 13: Data Gathering, Preprocessing & EDA

Problem Statement

You are working as a Junior Machine Learning Engineer. Before building any ML model, your responsibility is to collect data from multiple sources, clean and preprocess it, and perform Exploratory Data Analysis (EDA) to understand patterns, issues, and insights in the data.

This assignment focuses **ONLY** on pre-ML foundations:

1. Data gathering from different sources
2. Data preprocessing & cleaning
3. Exploratory Data Analysis (EDA)
 - ⚠ No model training in this assignment

Restrictions:

- Use Pandas, NumPy, Matplotlib, Seaborn, Requests, SQLite only.
- No ML model training yet.
- Code clarity is important.

PART 1 — Data Gathering (Multiple Sources)

Task 1: Load Data from CSV

1. Download any dataset from Kaggle (CSV format).
2. Load it using Pandas.
3. Print:
 - Shape of the dataset
 - Column names
 - First 5 rows
 - Other Preprocessing Techniques discussed in the lectures

Task 2: Load Data from JSON

1. Create or download a small JSON file (example: product data, user data).
2. Load it using Pandas.
3. Convert it into a DataFrame.
4. Print the DataFrame.

Task 3: Load Data from SQL Database

1. Create a SQLite database (sample.db)
2. Create a table named employees with columns:
 - id
 - name

- department
- salary
- 3. Insert at least 5 records.
- 4. Read the table into Pandas using SQL query.
- 5. Display the DataFrame.

Task 4: API Mini Project (TMDB API)

Mini Project: Movie Data Collector

1. Create a free account on TMDB (The Movie Database)
2. Generate an API key.
3. Using Python (requests library):
 - Fetch popular movies data
 - Extract fields:
 - Movie title
 - Release date
 - Rating
 - Popularity
 - Any Columns of Your Choice
4. Convert the API response into a Pandas DataFrame.
5. Save the collected data into a CSV file named tmdb_movies.csv.

Extra (optional): Fetch data for top-rated movies as well.

PART 2 — Data Preprocessing & Cleaning (Kaggle Dataset)

Use the same Kaggle dataset loaded in Task 1

Task 5: Understanding the Data

1. Check dataset shape.
2. Display column data types.
3. Identify numerical and categorical columns.
4. Check missing values count per column.
5. Rest of The Known Techniques

Task 6: Data Cleaning

Perform the following preprocessing steps:

1. Handle missing values:
 - Numerical → mean or median
 - Categorical → mode or "Unknown"
2. Remove duplicate rows (if any).

3. Rename columns to lowercase and snake_case.
4. Fix incorrect data types (example: string numbers to int/float) etc.

Task 7: Feature Preparation (Basic)

1. Convert categorical columns into numerical using:
 - Label Encoding OR
 - One-Hot Encoding
2. Separate features and target column (choose any relevant target).

PART 3 — Exploratory Data Analysis (EDA)

Task 8: Univariate Analysis

1. Plot distribution of numerical columns (histogram + KDE).
2. Count plot for categorical columns.
3. Identify outliers using boxplots.
4. Show Every Chart Possible being shown in matplotlib and Seaborn Lectures.

Task 9: Bivariate Analysis

1. Numerical vs Numerical:
 - Scatter plot
 - Correlation heatmap
2. Categorical vs Numerical:
 - Bar plot
 - Box plot
 - Show Every Chart Possible being shown in matplotlib and Seaborn Lectures for bivariate as well

Task 10: Insights & Observations

Write at least 5 insights, such as:

- Patterns
- Outliers
- Relationships
- Data quality issues

Write insights in markdown cells (if notebook) or comments.

Submission Guidelines

- Don't use AI
- Create ONE main folder.
- Example folder name: GenAI-Task(number)-YourName
- Inside this folder include: (All code files and Readme.md file must be inserted inside this folder)
- Convert the folder into ZIP format.
- Upload the ZIP folder on the submission portal.

If you have any doubt, you can directly contact the Tutedude mentor.